# UNIT 2: LASER AND APPLICATIONS



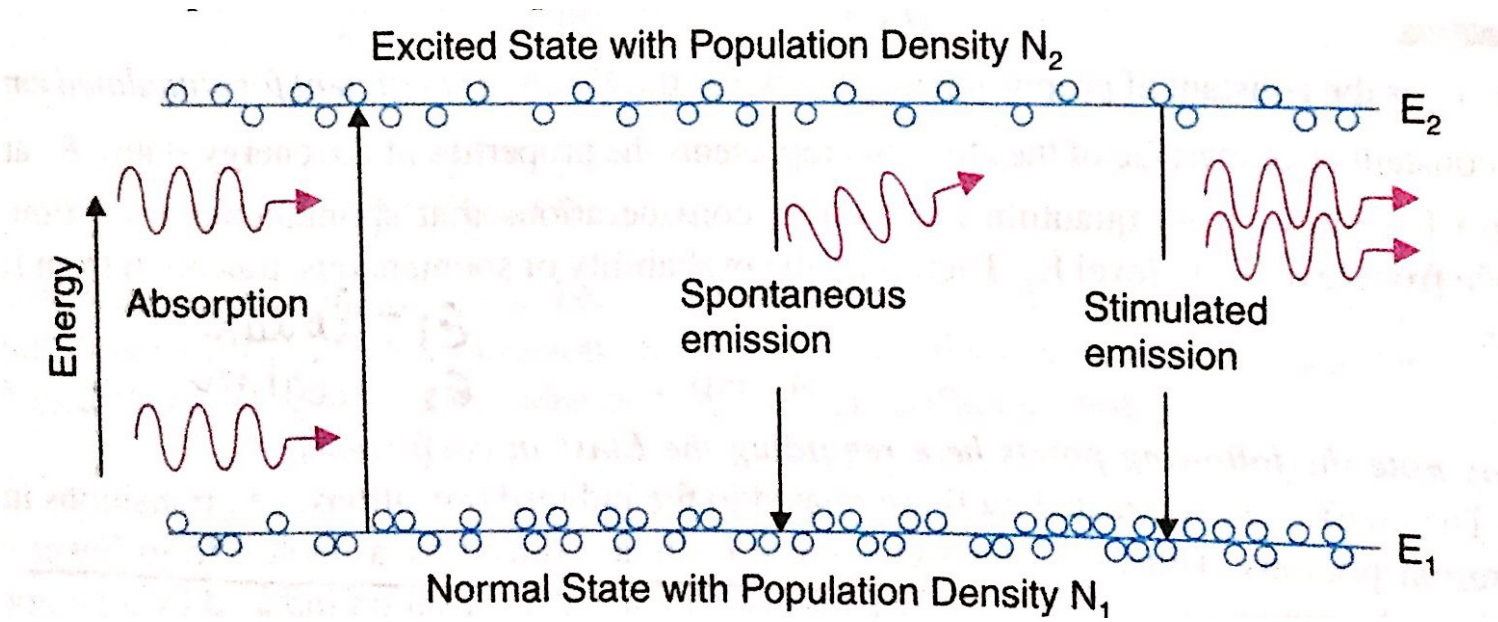
## LECTURE 4

# Revision Lecture 1



When a photon of energy (hν) travel through a material three different processes occurs

1. **Induced Absorption,** $R_{abs} = P_{12}N_1 = B_{12}\,\rho(\nu)N_1$
2. **Spontaneous Emission** $R_{sp} = P_{21}N_2 = A_{21}N_2$
3. **Stimulated Emission** $R_{st} = (P_{21})_{st}N_2 = B_{21}\,\rho(\nu)N_2$



# Revision Lecture 2



Populations of energy levels is given by Boltzmann's distribution

$$\frac{N_1}{N_2} = \frac{e^{-E_1/kT}}{e^{-E_2/kT}} = e^{(E_2-E_1)/kT}$$

$$R_{ab} = R_{sp} + R_{st}$$

**Absorption rate,** $R_{ab}$
**Stimulated Emission rate,** $R_{st}$
**Spontaneous Emission rate,** $R_{sp}$

$$B_{12}\,\rho(\nu)N_1 = A_{21}\,N_2 + B_{21}\,\rho(\nu)N_2$$

$$\rho(\nu) = \frac{A_{21}\,N_2}{[B_{12}\,N_1 - B_{21}N_2]}$$

$$\rho(\nu) = \frac{A_{21}}{B_{12}}\frac{1}{[e^{h\nu/kT} - B_{21}/B_{12}]}$$

Photon density (ρ)

$$\rho(\nu) = \left[\frac{8\pi h\nu^3\mu^3}{c^3}\right]\left[\frac{1}{[e^{h\nu/kT} - 1]}\right]$$

$$\frac{A_{21}}{B_{12}} = \frac{8\pi h\nu^3\mu^3}{c^3}$$

Planck's law

**Einstein relations**

$$\frac{B_{21}}{B_{12}} = 1$$

# Revision Lecture 2

$$R_1 = \frac{R_{st}}{R_{sp}} : \quad R_2 = \frac{R_{st}}{R_{abs}}$$

Ratios of stimulated emission rate to rate of spontaneous emission and rate of induced absorption will give us an Idea about what we need to realize LASER

To have LASER action dominant Stimulated emission is required

For that, we have to have high $R_1$ and $R_2$

$$R_2 = \frac{N_2}{N_1}$$

1. Population inversion

2. High photon density
3. Metastable state

$$R_1 = \frac{B_{21}}{A_{21}} \rho(\nu)$$

**1. POPULATION INVERSION by Pumping**

**2. METASTABLE STATE to increase life time**

**3. Large photon density (ρ) –Optical resonant cavity**

# Revision Lecture 3



1. Active medium- *metastable state to reduce* $A_{21}$
2. Pump- *population inversion to realize* $N_2 >> N_1$ **Components of LASER**
3. Optical resonant cavity- *confinement to increase* $\rho$

$$\gamma \geq \alpha_s + \frac{1}{2L} \ln \frac{1}{r_1 r_2}$$

**Known as the condition for Lasing**

$$\gamma_{th} = \alpha_s + \frac{1}{2L} \ln \frac{1}{r_1 r_2}$$

Eq.2

**Known as the threshold condition for commencing Lasing**

# Quick Quiz

The excitation mechanism in Laser is the mechanism to create population inversion. State True or False

a) True
b) False

**Ans: A**

Life time of atoms in the excited state is

a) $< 10^{-8}$ s
b) $> 10^{-6}$ s
c) $\sim 10^{6}$ s
d) $\sim 10$ s

**Ans: A**

Life time of atoms in the metastable state is

a) $< 10^{-8}$ s
b) between $10^{-6}$ -$10^{-3}$ s
c) ~$10^{6}$ s
d) ~10 s

**Ans: B**

# LECTURE SCHEDULE

**29/8/18 Lecture 1**:*Fundamentals of laser- energy levels in atoms,* **Radiation matter interaction, Absorption of light, Spontaneous emission of light, stimulated emission of light,** *Einstein A and B coefficients*

**30/8/18 Lecture 2: Einstein's relation, Population of energy levels, Metastable state, population inversion, Resonant cavity.**

***31/8/18 Lecture 3: Components of Laser, Excitation mechanisms, lasing action,***

**5/9/18 Lecture 4: *P*roperties of laser, Type of Lasers: Nd - YAG, He-Ne Laser, Semiconductor Laser,**

**6/9/18 Lecture 5: Applications of laser** in engineering, Holography.

**7/9/18 Lecture 6:**

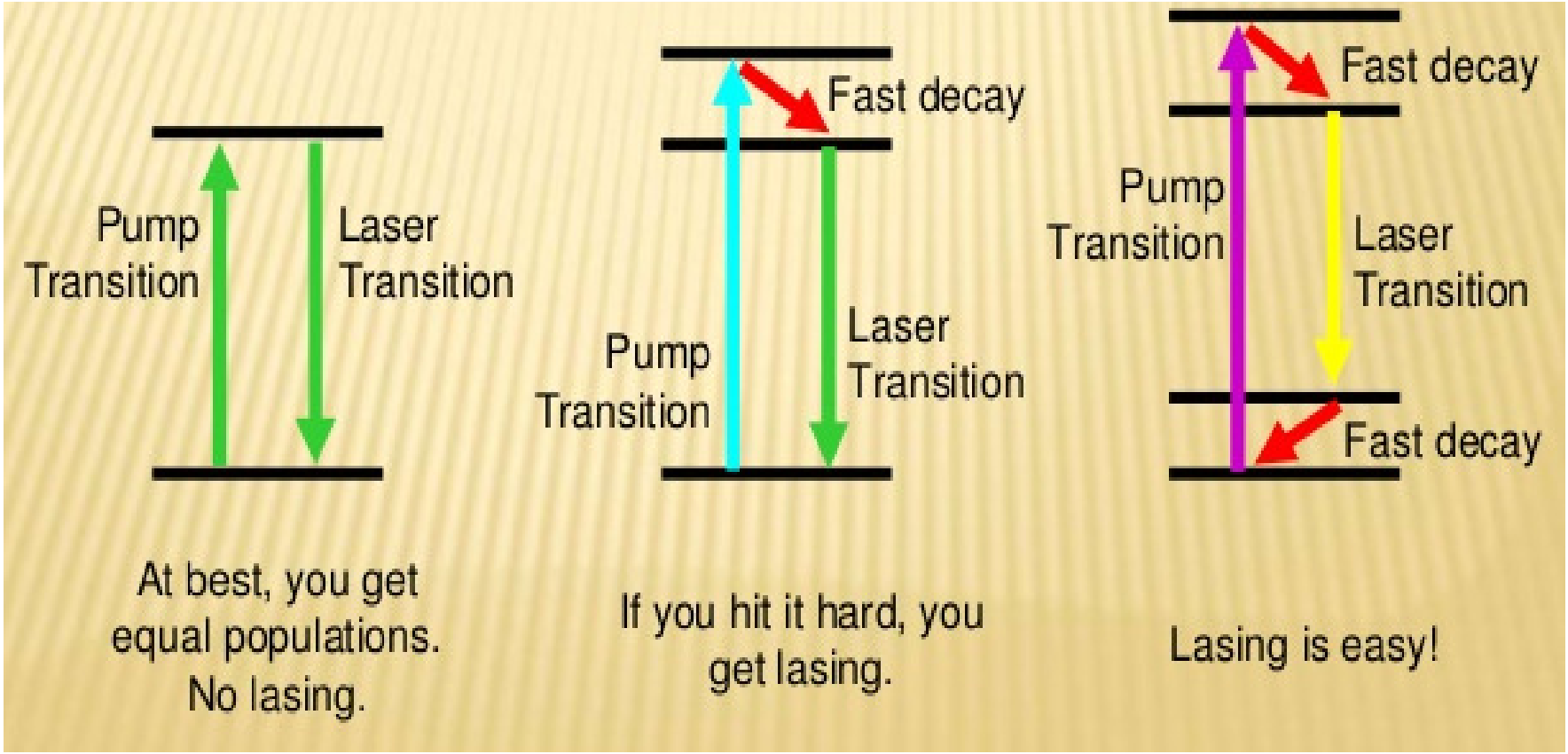
# PRINCIPAL PUMPING SCHEMES



## 2 Level LASER

## 3 Level LASER

## 4 Level LASER




**Not possible to get LASER**

**High pumping power**
**Pulsed laser**

**Low pumping power**
**Continuous laser**

# PROPERTIES OF LASER



Important characteristics of the laser beam's are

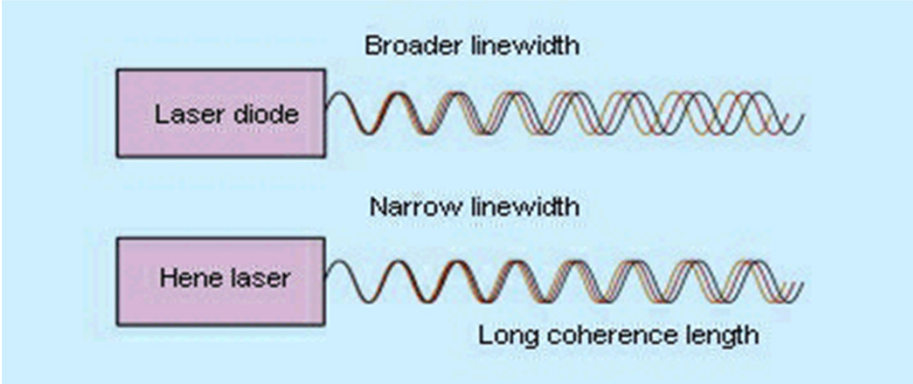
- **Coherence**
- **Monochromaticity**
- **Collimation**
  - **Directionality**
  - **Negligible divergence**
  - **High intensity**

# PROPERTIES OF LASER



## 1. Coherence

Identical photon emitted through stimulated emission are in **phase** with each other..




Responsible for amplification by interference….

Coherence length: Distance up to which phase remains same..Eg..

Na lamp ~ 0.3 mm

He-Ne Laser~100 mm

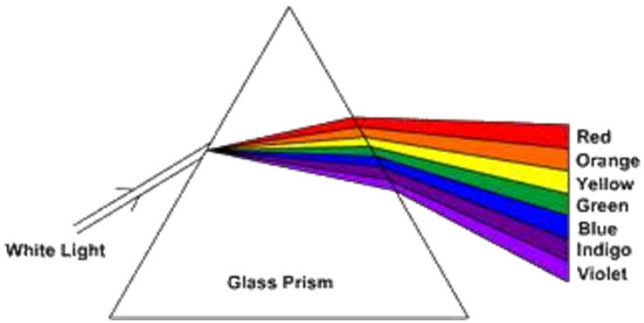
**Normal light source are highly incoherent.. Photons are out of phase with each other..**

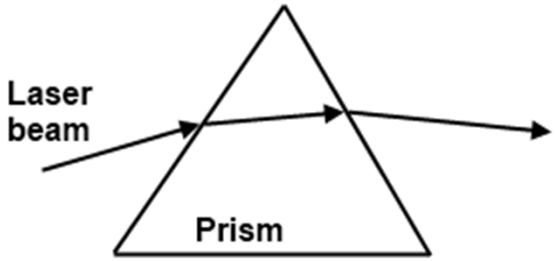
# PROPERTIES OF LASER



## 2. Monochromaticity

Electromagnetic radiation (light) of a single wavelength.




In traditional monochromatic source: variation in $\lambda(\Delta\lambda)$ ~ 100Å -1000Å
But with laser light source: : variation in $\lambda(\Delta\lambda)$ <10Å

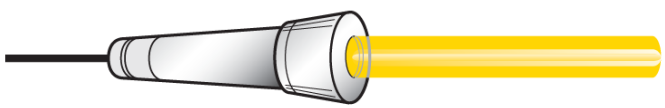
# PROPERTIES OF LASER

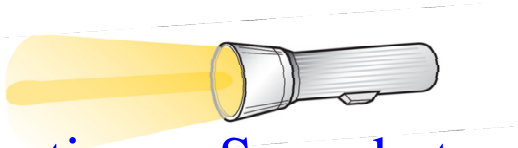


## 3. Collimation

### *3.1 Directionality*

- Cylindrical resonant cavity ensure light parallel to the cavity axis, so LASER light source emit light only in one direction



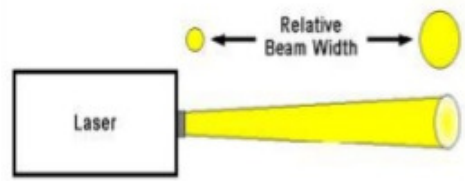


- But conventional light emit uniformly in all directions.. See what happens when we try to give it some direction by using lens and reflectors.. Like in torch

# PROPERTIES OF LASER



## *3.2 Negligible divergence*




Conventional sources spread light in the form of spherical wave front and hence highly divergent

LASER source light propagates in the form of plane wave front and hence very less divergent. If at all it exist it is due to the diffraction at the exit window.

Angle of divergence is given by

$$\phi = \frac{d_2 - d_1}{l_1 - l_2}$$

Where $d_1$ and $d_2$ are the spot diameters at $l_1$ and $l_2$, respectively./

# PROPERTIES OF LASER



### *3.3 High intensity*

Intensity of the light from LASER source almost remains constant but the intensity of light from conventional sources decreases rapidly with distance..

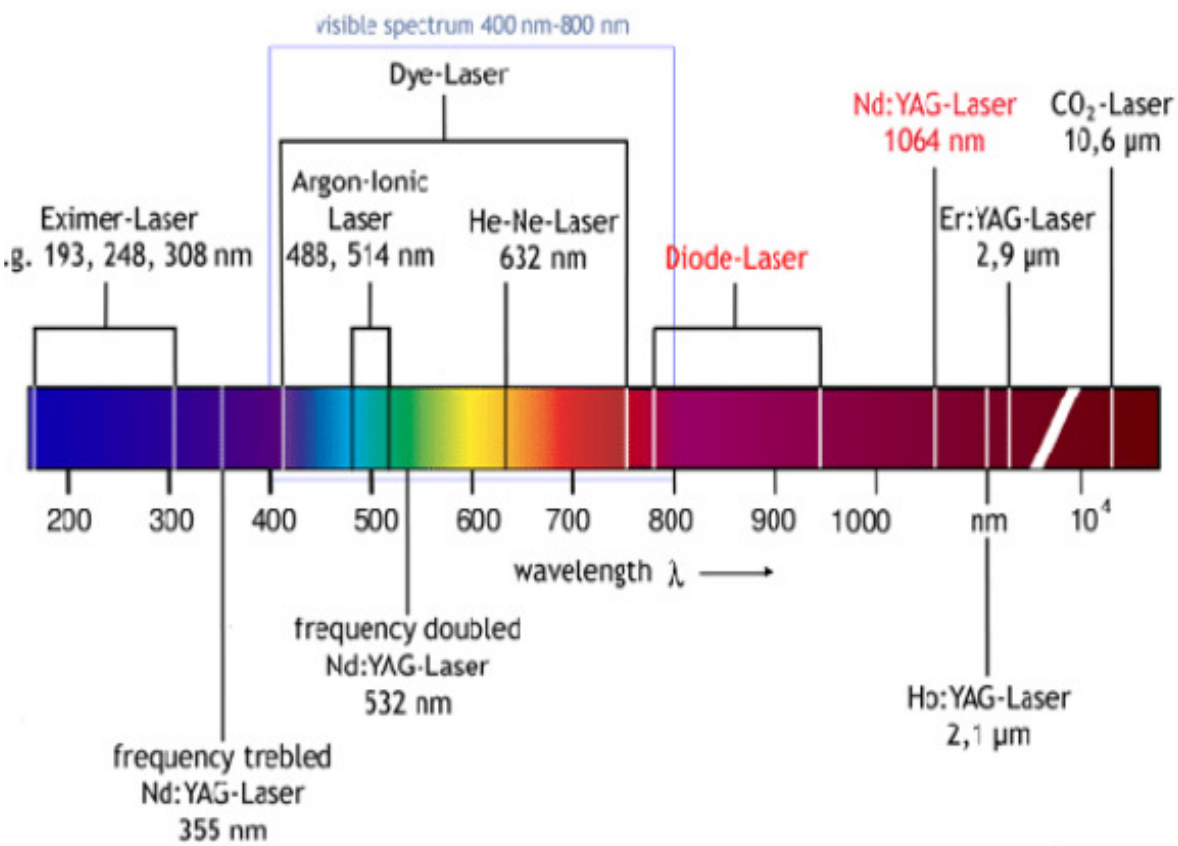
Intensity of the LASER beam is given by

$$I = \left(\frac{10}{\lambda}\right)^2 P$$

Where P is the power radiated by the laser and λ the wavelength

# TYPES OF LASERS: Classified on the basis of material used as active medium






1. Solid state laser
   a) Ruby laser (first laser)
   **b) Nd-YAG Laser**
2. Gas Laser
   a) $CO_2$ laser
   b) CO Laser
   c) Nitrogen laser
   d) Hydrogen laser
   **e)He-Ne Laser (first gas laser)**
3. Liquid laser
   Dye laser
4. Semiconductor laser
   **Laser diode- pn junction**

Wave length of the laser varies with the material used as the active center

# SOLID STATE LASER



**In a solid state laser, active material/active medium is solid material …either crystalline or amorphous (glassy)**

1. Ruby laser.
   i. First laser invented in 1960 by Theodore Maiman
   ii. Three level laser
   iii. Ruby crystal ($Al_2O_3$) with 0.05% Cr active centers
2. **Nd-YAG Laser**
   i. Most popular type of solid state laser
   ii. Four level Laser
   iii. Nd- $YAl_5O_{12}$

# SOLID STATE LASER



## ND:YAG LASER

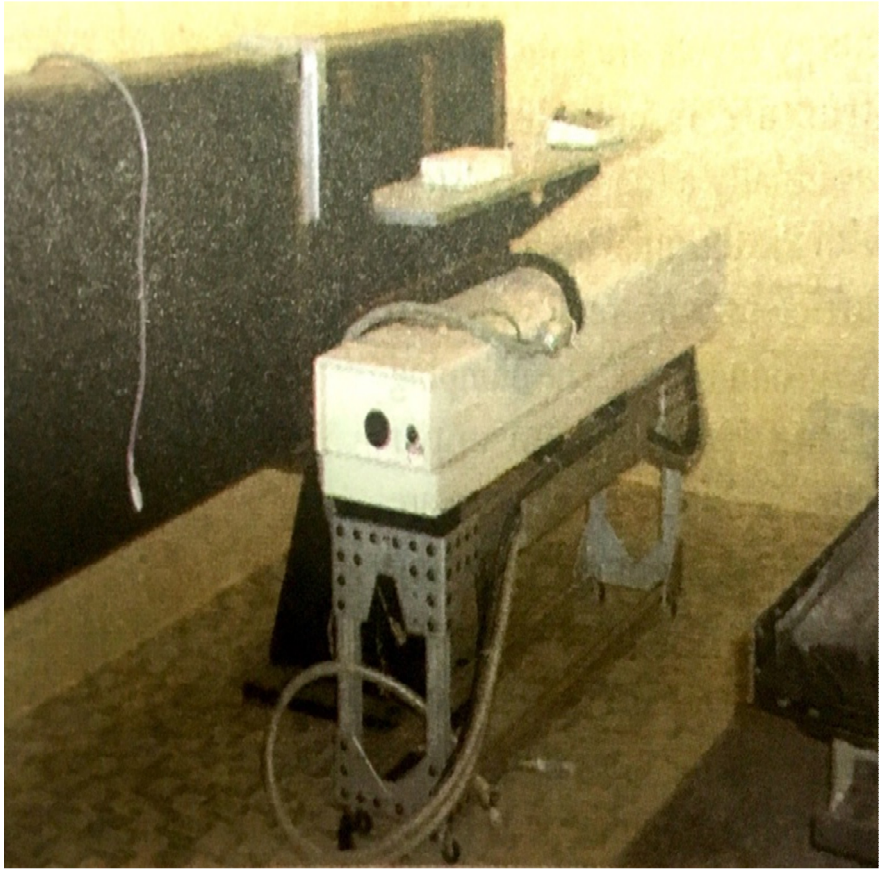
Operates in the IR region at a wavelength of about 1060 nm. Many industrial application such as welding, drilling , in surgery etc..

Yttrium aluminum garnet (YAG)– Host material $YAl_5O_{12}$

Neodymium (Nd).. Active centers..

$Nd^{3+}$ replace some $Al^{3+}$ ( 0.725% by wt)

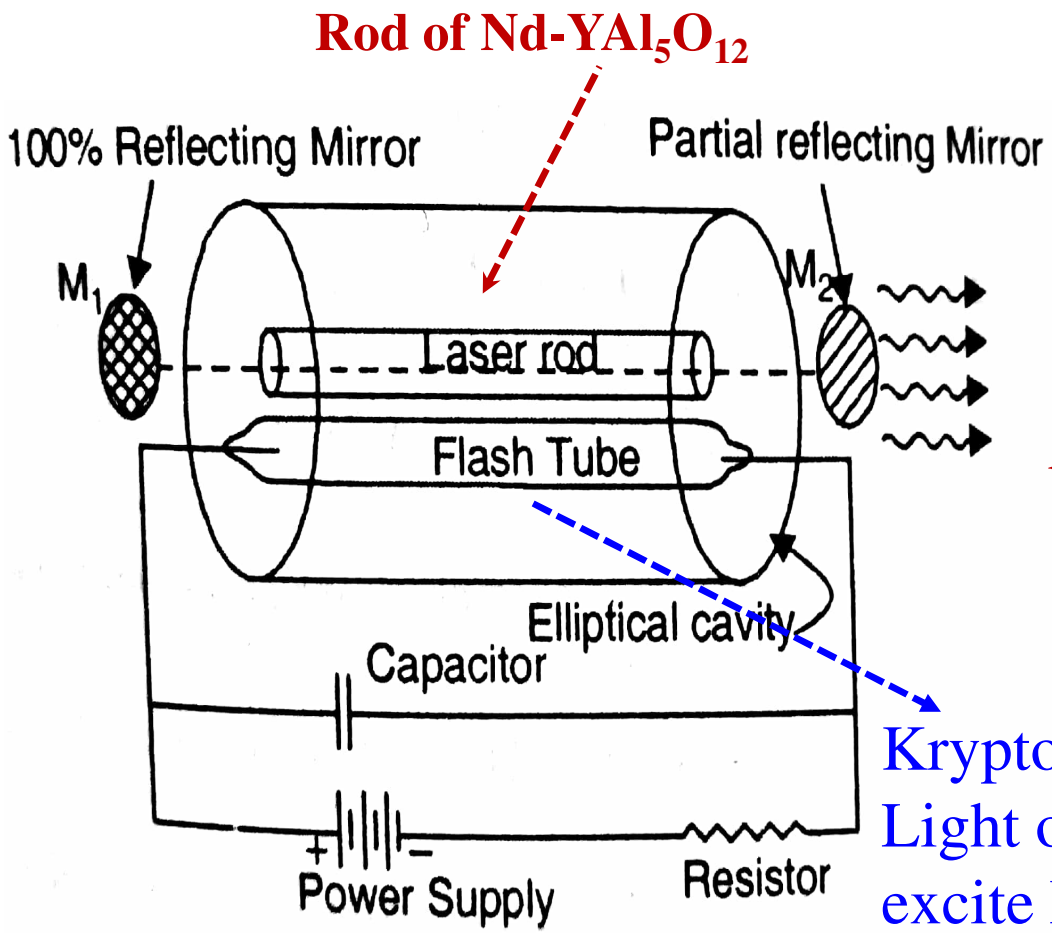
**Pumping source: Flash tube …light**



# SOLID STATE LASER



## Nd:YAG LASER- Design




Design of Nd-YAG Laser

a. Light from the flash tube reflected and fall entirely on the Nd-YAG rod.. That realize excitation by optical pumping and population inversion

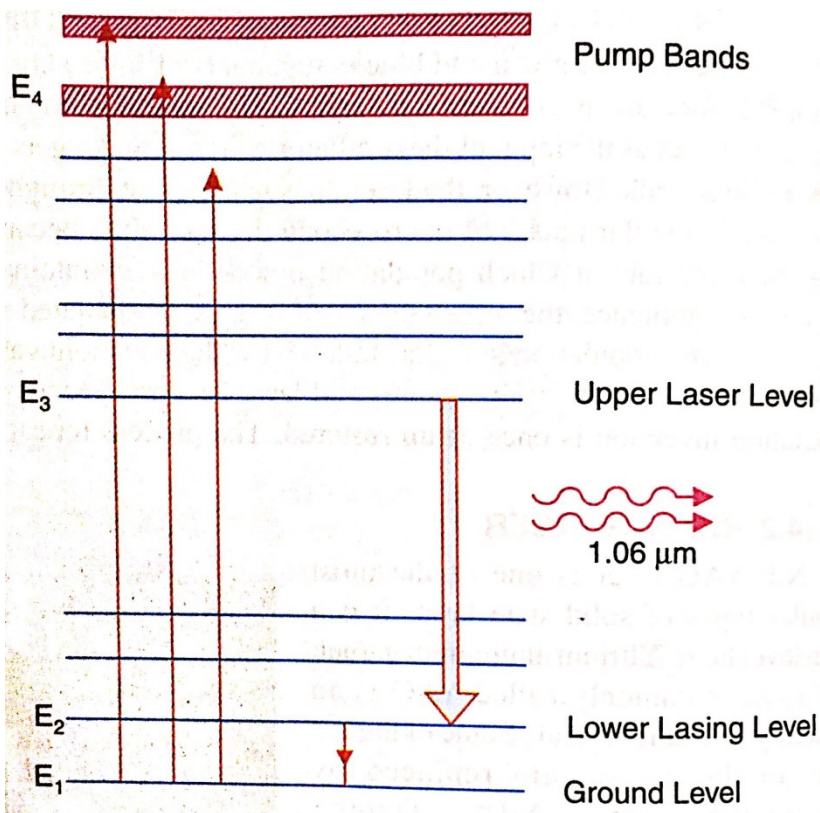
b. Two mirrors constitute the optical resonator..

Krypton Arc Lamp
Light of wavelength 500 to 800 nm excite $Nd^{3+}$ into higher levels

# SOLID STATE LASER



## Nd:YAG LASER




Energy levels and transitions in a Nd : YAG later.

1. $Nd^{3+}$ will be excited to $E_4$ levels from $E_1$ ground levels
2. $E_3$ is the metastable level and is the upper laser level.
3. $E_3$ will be rapidly populated by the optical pumping
4. $E_2$ is the lower leaser level and almost empty. Population inversion between $E_3$ and $E_2$.
5. Laser emission occurs in the Infrared region ( 1060 nm)
6. Used in welding, drilling and in surgery.

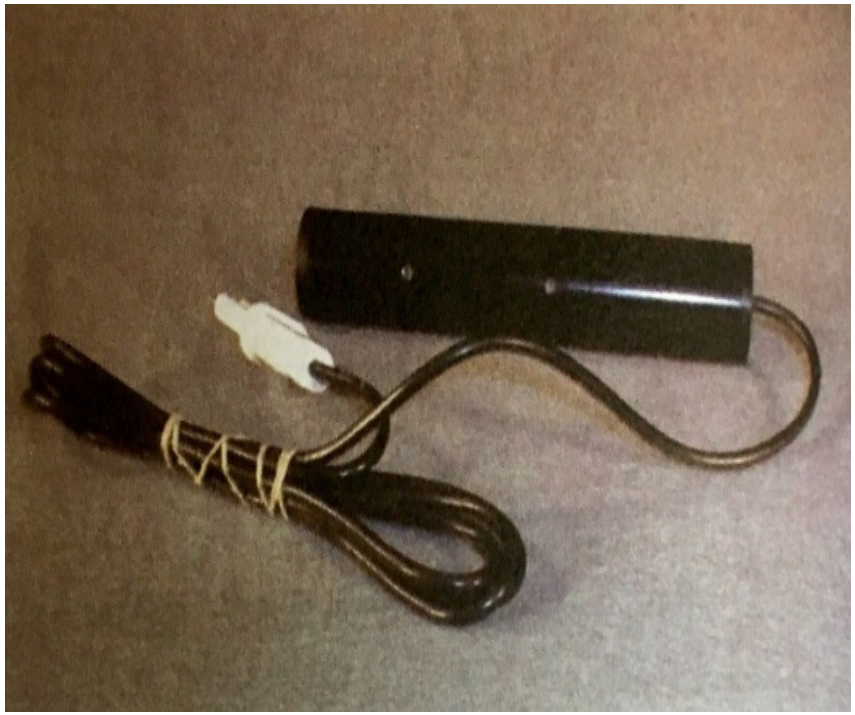
# GAS LASER



Gas Lasers are the most widely used lasers
Optical pumping is not used, but electric discharge used to excite atoms
Gas have narrow band and hence selection of optical source is difficult

## Helium-Neon LASER

- Operates in the visible region at a wavelength of about 632.8 nm. Operates in the CW mode widely used as monochromatic source in laboratories , in laser printing and bar-code reading..
- Invented in 1961 by Ali Javan, W.R. Bennet and D.R. Herriot

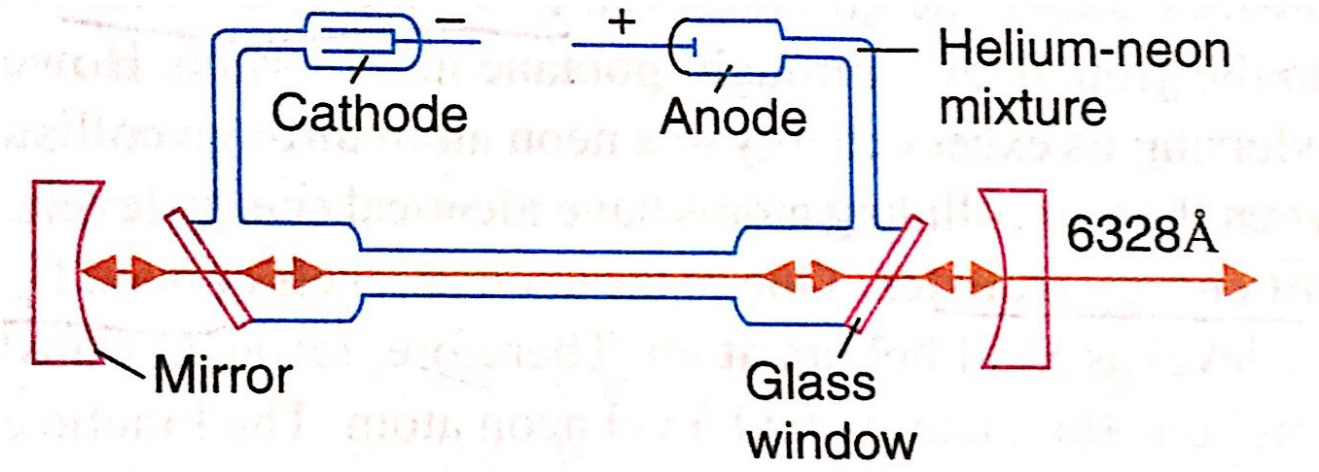


He-Ne gas laser is a low power laser.. Where as $CO_2$ gas laser is a high power laser used **for the industrial applications**

# GAS LASER



## Design of Helium-Neon LASER

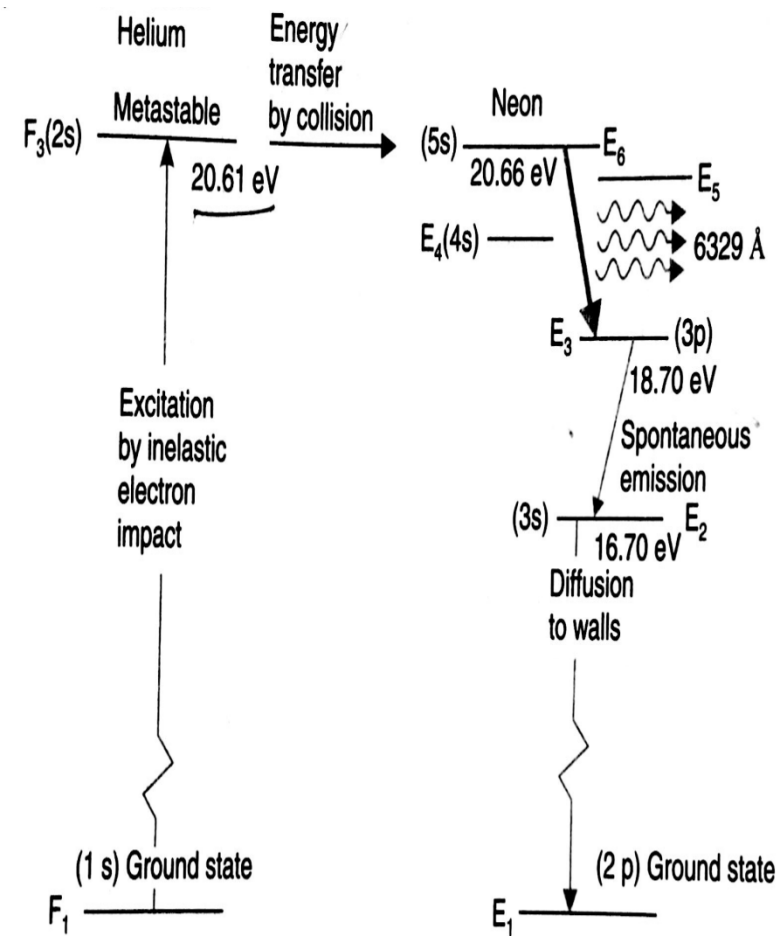



- ✓ He:Ne ratio is 10:1
- ✓ Neon atoms active centers
- ✓ He help to excite Ne atoms
- ✓ Pumping by electric discharge ~ 10 kV ionize gas:
- ✓ High energy electrons in the medium excite He atoms

# GAS LASER



## Working of Helium-Neon LASER




- Four level laser
- Excited He atom transfer energy to Ne atom and Ne act as active center
- $F_3$ of He is metastable state so no spontaneous emission. Return to ground state possible by energy transfer to Ne atoms
- resonant energy transfer- colliding atoms of identical energy level
- $E_2$ to $E_1$ by losing energy by collision with walls

# Semiconductor LASER



**LASER Diode: Portable, need only low input power and operate Operates in the IR to UV region. Widely used nowadays, in fiber optic communication, CD players, CR-ROM drives, high speed laser printing..**

1962- low temperature diode laser with GaAs
1970-Room temperature diode laser in CW mode

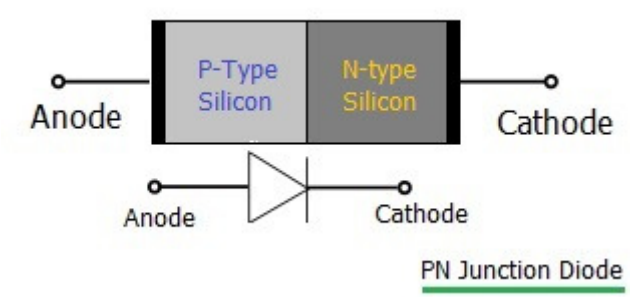
- Bands of solid are involved not discrete energy levels of atoms
- Holes and electrons combines to produce photon
- Lasing action occur at junction..

- ✓ Basically a p-n junction diode
- ✓ Small in size 0.1 mm long and hence portable
- ✓ Forward biased condition used
- ✓ Direct bandgap semiconductors are used

# Semiconductor LASER



P-n Junction diode




- Si is an indirect bandgap semiconductor.. And hence can not be used for Laser diode

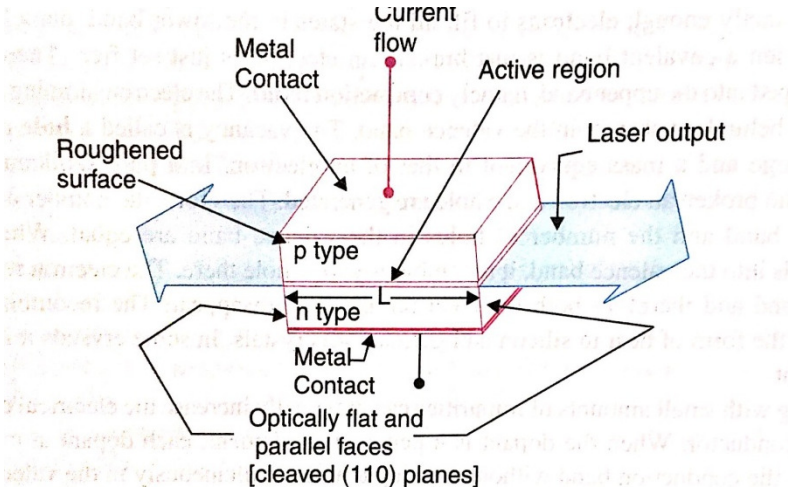
N-type, P-type, Fermi level
Band bending, Depletion region, junction????

Heavily doped?? We use this in laser diode..

# Semiconductor LASER



## Design of semiconductor Laser

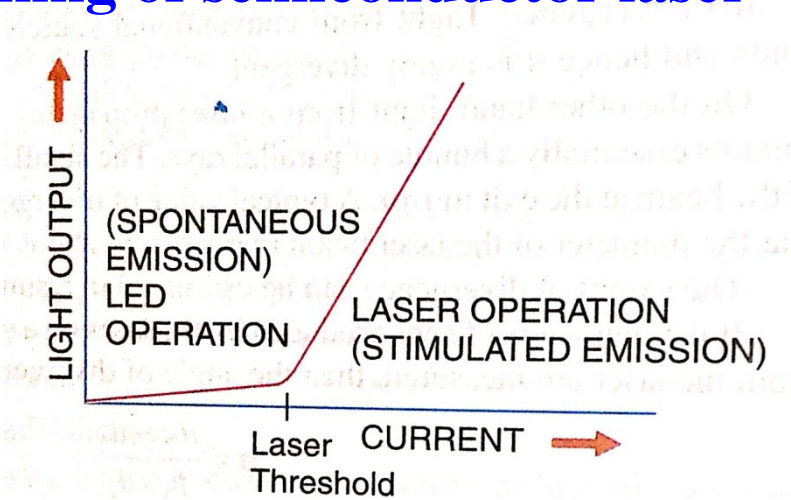



- ✓ GaAs on both side of the junction- direct band gap semiconductor
- ✓ Top and bottom faces metallized
- ✓ left and right polished.. Reflect and form optical resonator
- ✓ Front and back are roughened to avoid lasing action in that direction
- ✓ Entire structure packaged in a metal case and that look like a discrete transistor

# Semiconductor LASER



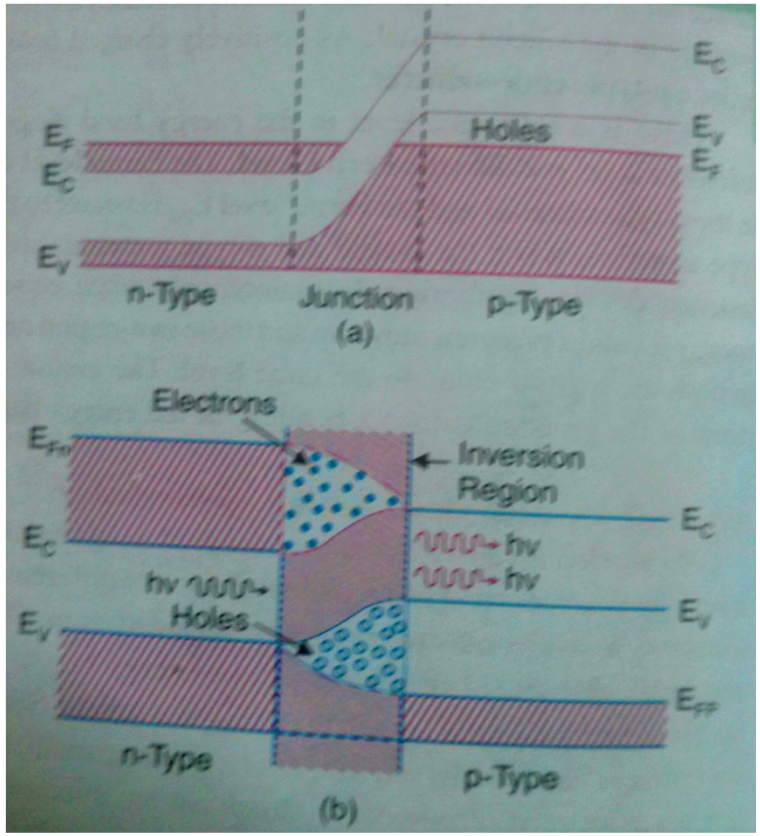
## Working of semiconductor laser




- Forward biased Jn. Electrons and holes injected into the junction region..
  Carriers are pumped by bias voltage
- Juction is the active region
- Electron hole recombination..

LED at low voltage
LASER at high voltage

Energy band diagram




With and without bias

# Some missed point about semiconductor laser

- ❑Depletion region
- ❑Charge injection to depletion region

- •Direct band gap semiconductor (GaAs)
- •Indirect band gap semiconductor (Si)

Gallium arsenide, indium phosphide, gallium antimonide and gallium nitride are all examples of compound semiconductor materials that can be used to create junction diodes that emit light

Si and Ge are indirect band gap material and hence not used

GaAS emit laser emit light 900 nm, IR region

# Quick Quiz

**Important characteristics of the laser beam's are**

a) Coherence
b) Monochromaticity
c) Collimation
d) All of the above
e) None of the above

In the following which is a three level laser system

a) Ruby Laser
b) $CO_2$ Laser
c) He-Ne Laser
d) Nd-YAG Laser

In the following which is a four level laser system

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